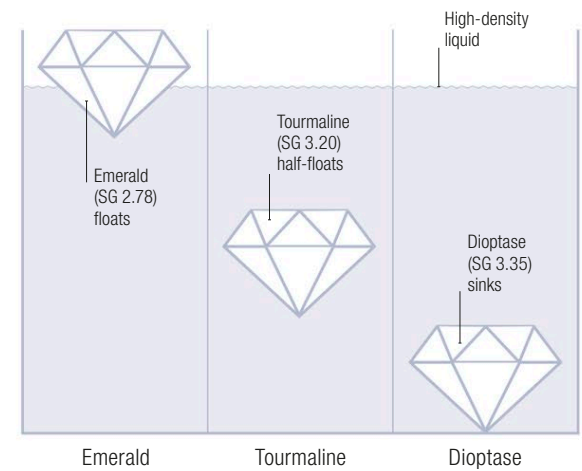




Aquamarine

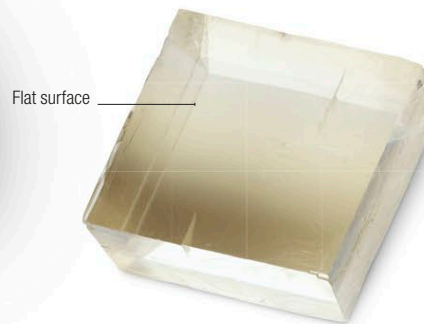
Specific gravity

Specific gravity (SG) is a measure of the density of a substance relative to that of water, and determines how dense a gemstone is. It is measured as the ratio of the mass of the substance and the mass of an equal volume of water – so a mineral with an SG of 2 is twice as heavy as water. Specific gravity can be determined using specialized balances or liquids that allow minerals below a given SG to float and those above it to sink. However, experts can often gauge the specific gravity of a gem purely by its heft.



Cleavage

Cleavage is the property of a mineral that causes it to break along its atomic layers, where the forces bonding its atoms are the weakest. Some gems have cleavages in several directions, some of which may be very easy to trigger, meaning that the gem can be easily broken if sharply knocked. Because they follow the atomic planes, cleavage surfaces are often smooth.



Perfect cleavage is the breakage of a mineral along an atomic plane, where the bonds are weakest, and where the breakage leaves a flat surface.

Streak

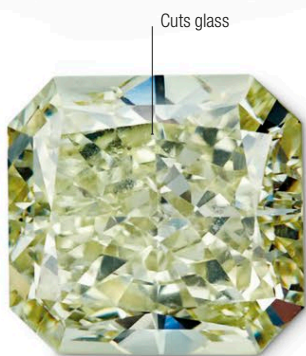
Streak is the colour of the powdered mineral, determined by rubbing a piece of the mineral across a piece of unglazed porcelain, leaving a streak of colour. This can be used for identification, particularly with minerals that occur in different colours.

Streak can be more consistent in minerals than colour – in fact, a mineral that occurs in different colour varieties may have the same colour streak. In this example, three colour varieties of the mineral fluorite rough are shown. In a streak test, the purple, orange, and green fluorite specimens would all give a white streak.



Fracture

Fracture is another way of describing how a mineral breaks. In fracture, however, breakage takes place across the mineral's atomic planes, rather than along them, as it does in cleavage, because there are no obvious planes of fracture. Distinctive fracture may help with identification.



Diamond

6000

7000

KNOOP SCALE (kg/mm²)

Obsidian

Conchoidal fracture is where the breakage has a shell-like appearance. Quartz and glass gemstones – such as obsidian – show conchoidal fracture.



Chalcopyrite

Even fracture has a broken surface that is roughly textured, but flat. **Uneven fracture**, as in the chalcopyrite above, has a rough and irregular surface.



Gold

Hackly fracture shows an uneven surface with sharp edges and jagged points. It is characteristic of broken or torn metals and a few other minerals.